Auto-classifier

The Auto-classifier action uses Regex to identify and extract codes, words and phrases in a document and store the result as a column value. Common uses are:

- Auto-classification
- Post migration clean-up
- Identifying content with e-discovery needed and discovery of content with privacy related information
- Retention and Disposal
- Other Business Processes based on document contents

What is Regex?

Regex (Regular Expressions) is a sequence of characters that define a search pattern. Regex patterns can be used to find and extract particular words, phrases, or values from any text. For this action you can use these patterns against the Name, Title and Contents of any documents. It is the contents search which makes this option particularly powerful.

If you are unfamiliar with Regex, then then there are a wide variety of resources available to help you with understanding and developing Regex patterns and Information Leadership Support can provide more information on these.

When searching contents, what document formats are supported?

Filename searches will work for all document types, but content searches will be limited to documents with the following extensions:

- *DOCX* – Microsoft Word XML
- *DOC* – Microsoft Word binary
- *TXT* – Plain text
- *RTF* – Rich text
- *PDF* – Text or OCR format. Image format not supported
- *MSG, EML* – Email

This support may be extended later if required.

PDF formats explained

PDF Text

- These store the document in a format very similar to a normal Word document
- They have the text complete with style and formatting information
- In Word, if you save the document as a PDF then it will be saved in this format
- If you open one of these documents and search, then that word will be correctly highlighted on the page

PDF OCR

- These are normally created by MFD's or scanners, though some other applications will also produce these when you choose 'Save as PDF'.
- With these each page is stored as single picture, normally in the TIFF format.
- The text is also stored separately without any styling, though some data on the texts position on the page is stored. When pages are opened the text is applied as a transparent layer over the image.
- If you open one of these documents and search, then the highlighting may or may not be positioned correctly depending on how accurately the transparent text layer matches the image.
- Often the text is not totally correct, between 75% to 95% accuracy with scanned documents.

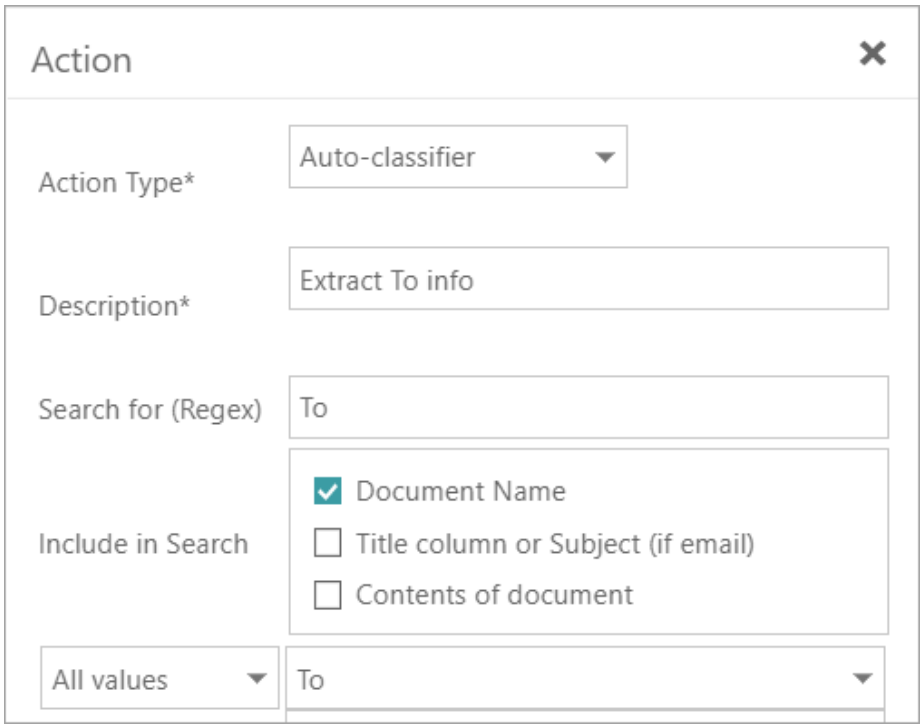
PDF Image

- These are also normally created by MFD's or scanners but where OCR has not been enabled.

- These store each page as a picture just like PDF OCR
- No text is stored
- You cannot search in one of these PDFs

Extraction of email related data

The **Auto-classifier** action supports some standard options to extract data from emails that have been saved to SharePoint as either eml or msg files.



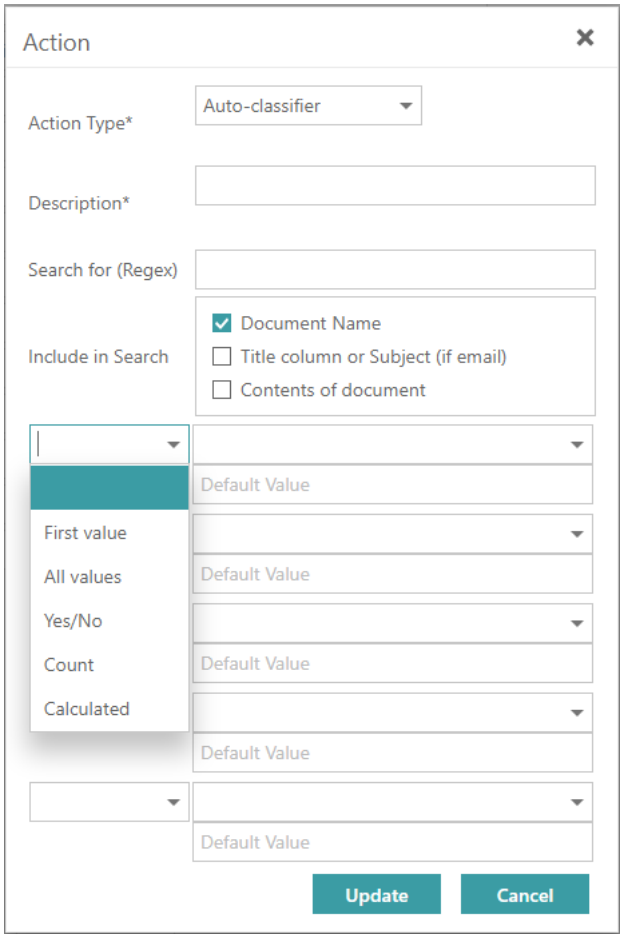
You can now enter any of the following into the **Search for (Regex)** field to extract the data from emails:

- To
- CC
- Bcc
- From
- Subject
- Sent

For To, CC and Bcc you also have the option to use the commands to extract different data:

- First value – returns the first person's email address
- All values – returns all email addresses as comma separated list
- Count – Count of the number of people

Configuring the Action



The action takes a regex pattern supplied by the user and searches the document for any matches. If matches are found, then one or more columns are updated. There should also be an option to update a column if no matches are found.

The action will have the following parameters:

Description

You can use this setting to enter a short description about the action. This has no impact on the action itself but is written to the aggregation logs when the rule is applied. It is useful to put in a description that is useful and understandable when viewed in the "edit rule" view.

Search for (RegEx)

The regex pattern to search for.

Include in Search

This defines where the regex pattern will search. By default, only Document Name is ticked but in most cases Contents of the document is also ticked. If you have emails and a separate 'Subject' column in your libraries for holding the Email subject, then you should tick Title column or Subject. If you do not have a separate subject column then the data extraction will attempt to check the subject while reading the contents.

As well as these initial settings you can also have up to five column update actions that use the data extracted to populate columns. Each of these column update actions have the following settings:

Column

The name of the column to update

Update Action Type

How to use the extracted data to populate the column, the options are:

- *First Value* – The first match found; all other matches are ignored. If more than one Search type option is ticked Filename will take precedence over Title and in turn Contents.
- *All values* – All of the matches in a comma separated list
- *Yes/No* – If one or more matches are found then set to Yes (true), otherwise No (false)
- *Count of found* – Count of the number of times the match was made
- *Calculated value* – Use a calculated value, options detailed below

Default Value

What value to set the column to if no matches are found. Automatically set to No if the action type is Yes/No

Value Calculation

This defines:

- *Normal Text* – Used as is
- *[Column Name]* – Value from another column
- *[First]* – First match found
- *[All]* – All matches found
- *[Count]* – Count of the number times the match was made
- *Blank* – Can be used when you want to ignore the found value but set the column to the default when the value has not been found

Recommended Column Types

Though a variety of column types can be used, these are the ones we recommend for the different update action types:

- *First Value* – Single or Multi-line text (set to plain text)
- *All Values* – Multi-line text (set to plain text). Single line text can be used but if values exceed 255 characters then it will be truncated.
- *Yes/No* – Yes/No column - other columns will display true or false instead of yes or no
- *Count of Found* – Numeric or Single line text
- *Calculated Value* – Multi-line text (set to plain text). Single line text can be used but if value exceeds 255 characters then it will be truncated.

Email Data Extraction

The Auto-classifier action can also be used to extract data from emails that have been saved to SharePoint as either eml or msg files.

You must select the “Contents of the Document” when extracting email metadata.

You can now enter any of the following into the Search for (Regex) field to extract the data from emails:

- To
- CC
- Bcc
- From
- Subject
- Sent

For To, CC and Bcc you also have the option to use the commands to extract different data:

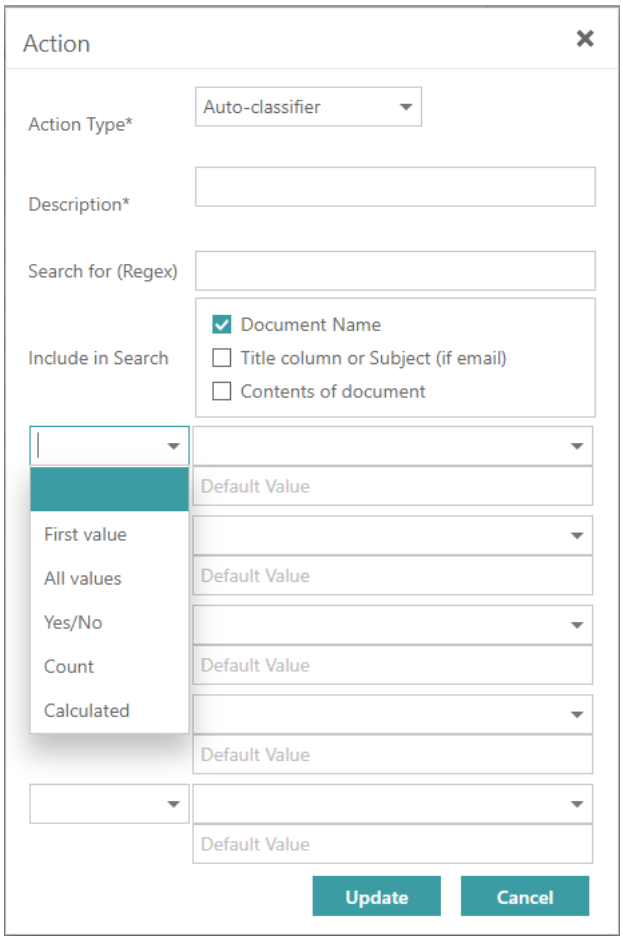
- First value – returns the first person’s email address
- All values – returns all email addresses as comma separated list
- Count – Count of the number of people

Example of a Data Extraction Action

If we wanted to extract references from documents where the reference started with a D followed by six numbers, then the regex would look like this:

```
(D\d{6})(?:\w\s)?
```

If we wanted to populate five columns (PRA Text 1 to 5) with the results of each type of update action type, then the resulting action would look like this:



If we ran this rule, then we would see updates like this against the documents in our library:

Name	PRA Text 1	PRA Text 2	PRA Text 3	PRA Text 4	PRA Text 5	Subject	Title
Annual Inspection.pdf	D567890	D123456,D567890	True	2	First is D567890 in 2 items D123456,D567890		
Approval for D123456.docx	D123456	D123456,D987654,D666666,D113355	True	4	First is D123456 in 4 items D123456,D987654,D666666,D113355		Test Title D113355
RE WLSP1 D112233.msg	D112233	D112233,D888888,D225599	True	3	First is D112233 in 3 items D112233,D888888,D225599	Test Subject D225599	